

Roll No.

TEE-301

B. TECH. (EE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
ELECTRICAL CIRCUIT ANALYSIS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Explain KCL and Supernode. Find the power delivered by the 5 A current source in the network shown in Fig. 1 (a). (CO1, CO2)

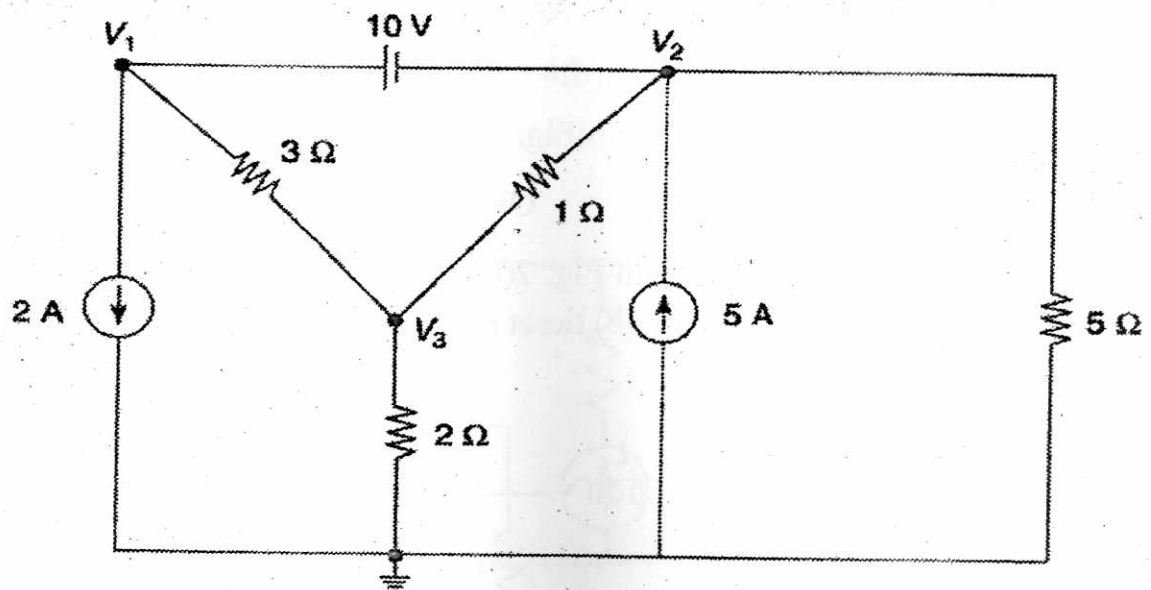


Fig. 1 (a)

P. T. O.

(2)

OR

- (b) Find V_{Th} and R_{Th} between terminals A and B of the network shown in Fig. 1 (b). (CO1, CO2)

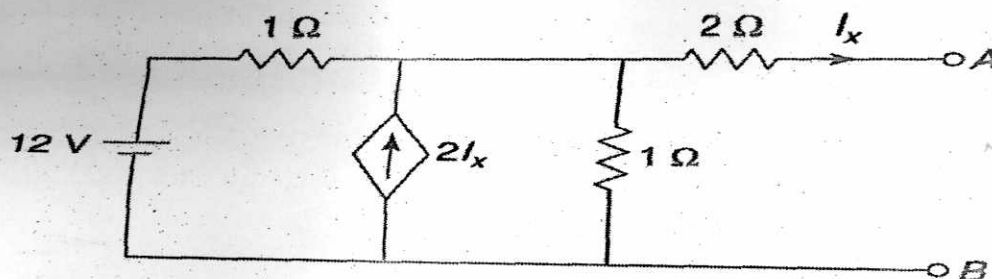


Fig. 1 (b)

2. (a) Fig. 2(a) shown a graph of the network. Show all the trees of the graph. (CO3)

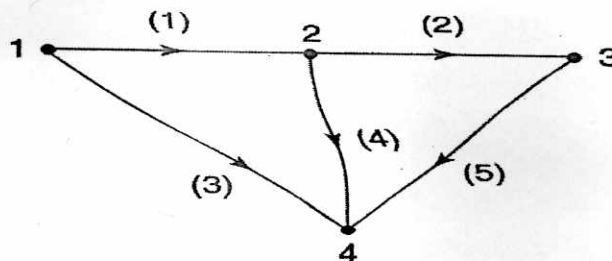


Fig. 2 (a)

OR

- (b) For the circuit shown in Fig. 2(b), draw the oriented graph and write the (a) incident matrix, (b) tieset matrix. (CO3)

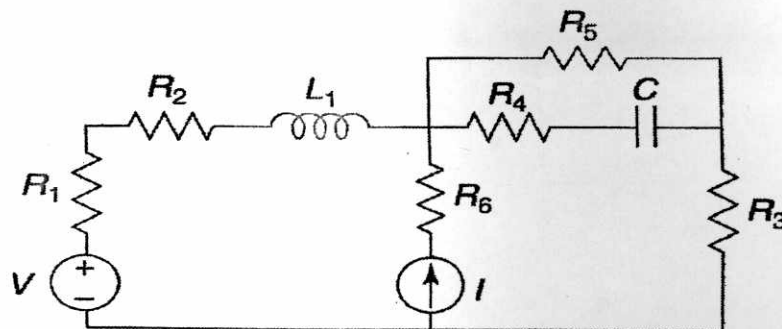


Fig. 2 (b)

3. (a) When a sinusoidal voltage of 120 V is applied to a series RL circuit, it is found that there occurs a power dissipation of 1200 W and a current flow given by $i(t) = 28.3 \sin (314 - \phi)$. Find the circuit resistance and inductance. (CO2, CO3)

OR

- (b) Three equal impedances, each of $(8 + j10)$ ohms, are connected in star. This is further connected to a 440 V, 50 Hz, three-phase supply. Calculate (i) phase voltage, (ii) phase angle, (iii) phase current, (iv) line current, (v) active power, and (vi) reactive power. (CO2, CO3)
4. (a) A series resonant circuit has an impedance of 500 ohm at resonance frequency. Cut-off frequencies are 10 kHz and 100 Hz. Determine (i) resonance frequency, (ii) value of L, C, and (iii) quality factor at resonance frequency. (CO2, CO3)

OR

- (b) For the network shown in Fig. 4(b), find the value of R_L for maximum power transfer. Also, find the maximum power. (CO1, CO2)

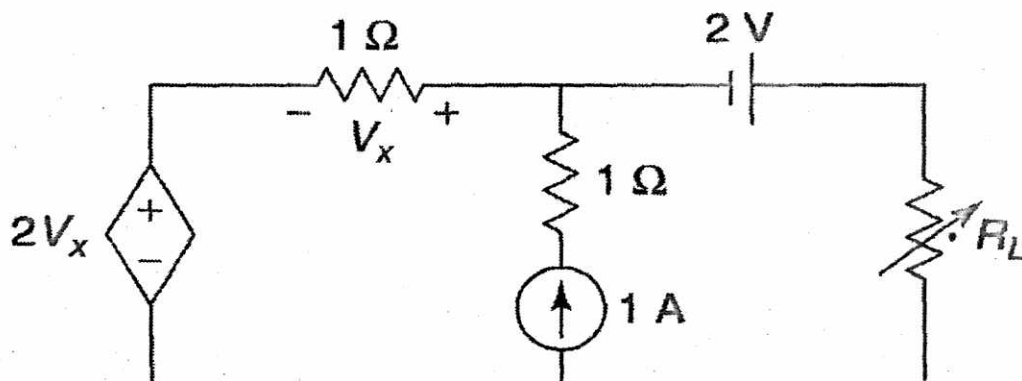


Fig. 4 (b)

5. (a) Find the Norton's equivalent network of Fig. 5(a). (CO1, CO2)

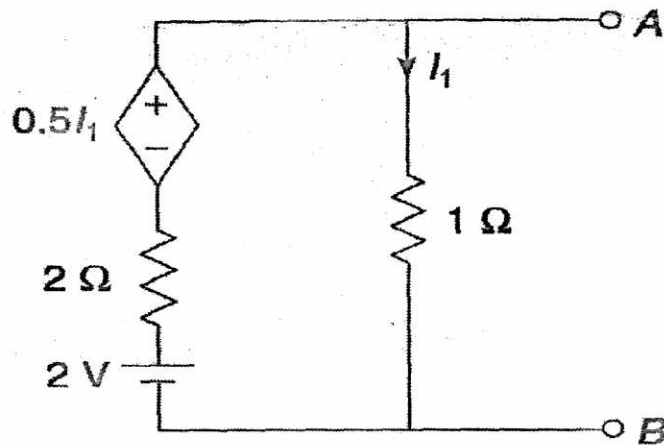


Fig. 5 (a)

OR

- (b) Find the Thevenin equivalent network of the network to the left of A-B in the Fig. 5(b). (CO1, CO2)

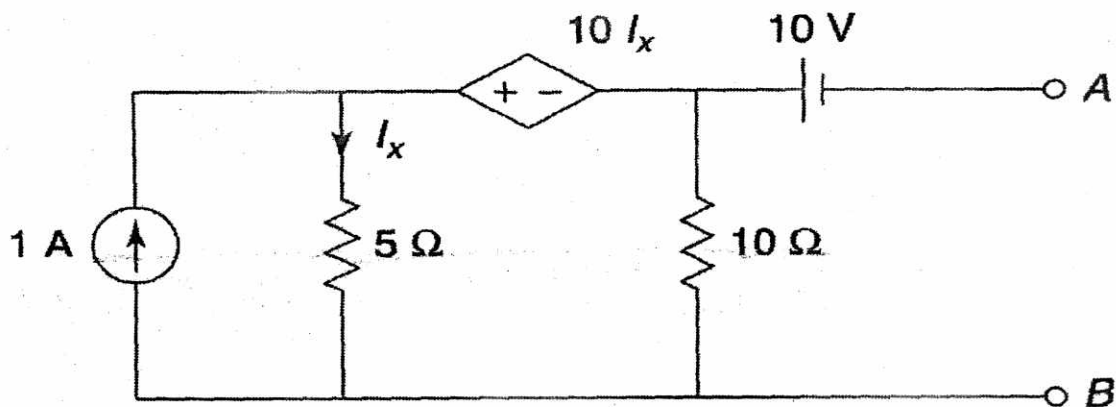


Fig. 5 (b)

Roll No.

TEE-302

**B. TECH. (EE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023**

ELECTRICAL MACHINE-I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the question by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Compare between electrical and magnetic circuits. (CO3)

OR

- (b) The flux in a magnetic core is alternating sinusoidally at a frequency of 600 Hz. The maximum flux density is 2 T and the eddy-current loss is 15 W. Find the eddy-current loss in the core if the frequency is raised to 800 Hz and the maximum flux density is reduced to 1.5 T. (CO3)

2. (a) Discuss hysteresis and eddy current loss with suitable diagram. (CO2)

OR

- (b) The core-loss (hysteresis + eddy-current loss) for a given specimen of magnetic material is found to be 2000 W at 50 Hz. Keeping the flux density constant, the frequency of the supply is raised to 75 Hz resulting in core-loss of 3200 W. Compute separately hysteresis and eddy-current losses at both the frequencies. (CO2)

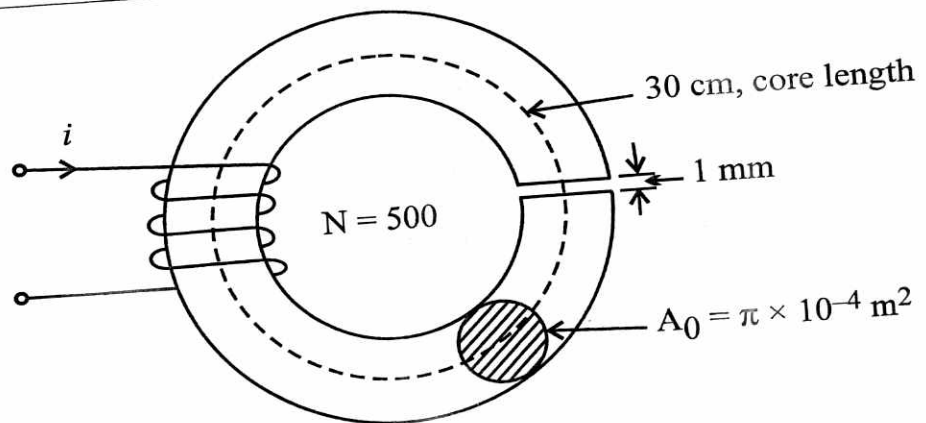
P. T. O.

(2)

3. (a) A wrought iron bar 30 cm long and 2 cm in diameter is bent into a circular shape as shown below. It is then wound with 600 turns of wire. Calculate the current required to produce a flux of 0.5 mWb in the magnetic circuit in the following cases : (CO1)

- (i) no air-gap :
 (ii) with an air-gap of 1 mm; $\mu_{r(\text{iron})} = 4000$ (assumed constant); and
 (iii) with an air-gap of 1 mm; assume the following data for the magnetization of iron :

H in At/m	B in T
2500	1.55
3000	1.59
3500	1.6
4000	1.615



OR

- (b) Explain energy balance equation for the operation of motor and generator. (CO1)

(3)

4. (a) Deduce the relation between energy and co-energy when armature is stationary. (CO1)

OR

- (b) Deduce the relation between energy and co-energy when armature is displaced. (CO1)
5. (a) Deduce the equation of torque for singly excited system. (CO4)

OR

- (b) Deduce the equation of torque for multiply excited system.

Roll No.

TEE-303

B. TECH. (EE) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

**ELECTRICAL AND ELECTRONICS MEASURING INSTRUMENTS
(EMMI)**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain indirect method of measurement. (CO1)

OR

- (b) What do you mean by static correction ? A meter reads 127.50 V and the true value of the voltage is 127.43 V. (CO1)

Determine :

- (i) The static error
(ii) The static correction for this instrument.
2. (a) What are static characteristics of instruments ? (CO1)

OR

- (b) What is measurement ? Why is it important ? (CO1)

P. T. O.

(2)

3. (a) Explain the following : (CO2)

- (i) Deflecting Torque
- (ii) Controlling Torque
- (iii) Damping Torque
- (iv) Power factor

OR

(b) Give the working principle of Moving Iron (MI) instruments. (CO2)

4. (a) What is multi-range Ammeter ? Explain in detail. (CO3)

OR

(b) Critically comment on Power in three phase system. (CO3)

5. (a) What are the different methods of measuring low resistances ? Explain them in brief. (CO3)

A highly sensitive galvanometer can detect a current as low as 0.1 nA . This galvanometer is used in a Wheatstone bridge as a detector. The resistance of galvanometer is negligible. Each arm of the bridge has a resistance of $1\text{ k}\Omega$. The input voltage applied to the bridge is 20 V . Calculate the smallest change in resistance which can be detected. The resistance of the galvanometer can be neglected as compared with the internal resistance of bridge.

OR

(b) Explain Maxwell's Inductance AC bridge with phasor diagram and mathematical analysis. (CO3)

Roll No.

TEE 304

**B. TECH. (EE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
ELECTROMAGNETIC FIELDS**

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) If the co-ordinate of any point in Cartesian co-ordinate system is (x, y, z) then find the co-ordinate in spherical and cylindrical co-ordinate system with suitable diagram. (CO1)

OR

- (b) Derive the relation between unit vectors in spherical and Cartesian co-ordinate systems. (CO1)
2. (a) Discuss the physical interpretation of curl, divergence and gradient. (CO1)

OR

- (b) If the variation of scalar field ϕ with space is given as : (CO1)

(i) $\phi(x, y, z) = 3x^2y - y^3z^2$

(ii) $\alpha(x, y, z) = 3x^2y + -y^3 + z^2$

Find the gradient of $\phi(x, y, z)$, and $\alpha(x, y, z)$.

P. T. O.

(2)

3. (a) Discuss divergence theorem.

(CO1)

OR

- (b) Find the angle between the surface $x^2 + y^2 + z^2 = 9$; and $x^2 + y^2 - z = 3$.

(CO1)

4. (a) Discuss Coulomb's law, electrical field intensity, and electric field potential.

(CO2)

OR

- (b) Point charge 1 mC and -2 mC are located at (3,2,-1) and (-1,-1,4), respectively. Calculate the electric force on a 10 nC charge located at (0,3,1) and electric field intensity at that point.

(CO3)

5. (a) Discuss electrical flux density and Gauss' law.

(CO2)

OR

- (b) Using Gauss' law find electric field intensity at a point due to infinite sheet of charge.

(CO3)

Roll No.

TEE-305

**B. TECH. (EE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023**

ANALOG ELECTRONICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) If a pure silicon crystal has 1 million free electrons inside it, how many holes does it have ?

What happened to the numbers of free electrons and holes if the ambient temperature increases ? (CO1)

OR

- (b) Explain that the current in a semiconductor is due to the combined effect of :

(i) Flow of free electrons, and

(ii) The flow of holes (CO1)

2. (a) Explain working of a half wave rectifier circuit. Draw wave shapes. Label fully the diagrams drawn. (CO1)

P. T. O.

(2)

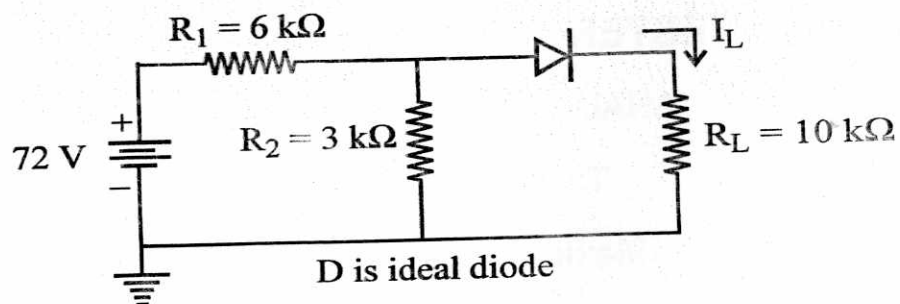
TEE-305

OR

(b) Differentiate between transition capacitance and diffusion capacitance.

What is a varactor diode and its doping pattern ? (CO1)

3. (a) Calculate the magnitude of current I_L through the load resistance R_L of 10k ohm in the following circuit : (CO1)



OR

(b) An N-P-N transistor is connected in common Emitter configuration. Explain and draw the following static characteristics :

(i) Input characteristic.

(ii) Output or collector characteristics.

(iii) Current transfer characteristics. (CO1)

4. (a) Draw various types of circuit configurations for operating a transistor. Label all the currents and voltages in the circuits.

Write characteristic parameters of any one of the configurations. (CO2)

OR

(b) Write the working of a JFET. Describe with purpose the following notations :

(i) Source

(3)

(ii) Drain

(iii) Gate

(iv) Channel

(CO2)

5. (a) Draw the symbols of the following. Label all the terminals with terminal voltages.

(i) N- Channel JFET

(ii) P- Channel JFET

(iii) N- Channel DE MOSFET

(iv) P- Channel DE MOSFET

(v) N- Channel E-only MOSFET

(vi) P- Channel E-only MOSFET

(CO2)

OR

(b) Draw and explain the static characteristics of a DE MOSFET. (CO2)

Roll No.

VAC-301

B. TECH. (EE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

BASICS OF MATLAB

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) What is Simulink ? Elaborate its application in electrical engineering.

(CO1)

OR

- (b) Generate 3×3 matrix using the help of MATLAB. Write the command for determination of (i) both diagonals of matrix (ii) inverse of the matrix (iii) rank of the matrix (iv) eigenvalues (v) sum of its column (vi) sum of its row.

(CO1)

2. (a) What do you mean by Array ? Explain Numeric Array and Character Arrays with suitable example.

(CO1)

OR

- (b) Write a MATLAB program to determine RMS value of sampled voltage signals. Use 256 samples per cycle and total six cycles.

(CO1)

P. T. O.

(2)

3. (a) What is state flow ? Discuss the type of state flow and its application.

(CO2)

OR

- (b) Generate a matrix :

(CO2)

$$A = \begin{bmatrix} 16 & 3 & 2 \\ 5 & 10 & 11 \\ 9 & 6 & 7 \end{bmatrix}$$

What will be the output of the following commands ?

- (i) `sum(diag(A))`
 - (ii) `sum(diag(fliplr(A)))`
 - (iii) `A(1, 2) + A(3, 3) + A'(2, 2)`
 - (iv) `sum(A(1:3, 2)) + sum(A(1:3, 3))`
4. (a) Discuss Mathematical expressions, (i) variables (ii) numbers (iii) operators, and (iv) functions, used in MATLAB. (CO3)

OR

- (b) In d.c. generator, the e.m.f generated due to rotation of the armature is given as : (CO3)

$$E = \frac{\phi Z N}{60} \times \frac{P}{A}$$

Consider $P = A$ and $\phi = 27 \text{ mWb}$, write a MATLAB program to draw plot between E and N . Here, $E = \text{e.m.f. generated}$, $Z = \text{total number of armature conductor}$, $N = \text{speed in r.p.m.}$

5. (a) Discuss the operators (i) Arithmetic operator (ii) Relation operators (iii) Logical operators (iv) Short circuit operator used in MATLAB.

(CO3)

OR

- (b) Discuss the application of (i) Bus selector (ii) Bus creator (iii) Scope block used in MATLAB. (CO3)